PRODUCT REQUIREMENTS DOCUMENT

RiskHub

A Proactive Web3 Cross-Exchange Risk Management and Behavioral Identity Platform

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1. Executive Summary

RiskHub is a proactive, read-only Web3 portfolio risk management platform designed to serve crypto investors operating across multiple centralized exchanges. In a market defined by emotional volatility, opaque performance claims, and fragmented asset visibility, RiskHub positions itself not as a trading tool, but as a behavioral and analytical shield — a platform that observes, quantifies, and warns.

The platform solves three compounding problems that disproportionately harm retail traders: (1) the inability to see one's full cross-exchange risk exposure in a single view; (2) the lack of an objective feedback mechanism to identify and interrupt emotional trading behaviors; and (3) the absence of tamper-proof, verifiable on-chain credentials for trading performance, which enables fraudulent copy-trade schemes.

RiskHub addresses each of these through four integrated core capabilities: cross-exchange read-only data ingestion via the CCXT library, a real-time Python-based Behavioral Quant Engine, an interactive Asset Contagion Graph, and an immutable Web3 Soulbound Token (SBT) that permanently encodes a user's verified trading discipline on-chain.

The MVP scope targets active crypto traders on Binance and OKX, delivering a functional dashboard, behavioral alerting system, and SBT minting on BNB Chain Testnet within a 16-week build cycle.

Attribute	Detail	Attribute	Detail
Product Name	RiskHub	**Lead Author**	Product Management Team
Version	1.0 (MVP)	**Stakeholders**	Product, Engineering, Design, Legal
Target Launch	Q4 2025	**Review Cycle**	Bi-weekly Sprint Review

2. Problem Statement

The cryptocurrency trading ecosystem imposes asymmetric information disadvantages on retail participants. Three structural problems converge to create an environment where retail traders consistently underperform and are actively exploited.

2.1 Fragmented Cross-Exchange Portfolio Visibility

Professional and retail traders alike routinely operate across 3–10 exchanges simultaneously to access diverse liquidity pools, hedging instruments, and yield products. No native tooling aggregates Spot and Futures balances — including unrealized PnL, locked margin, and liquidation thresholds — into a single, coherent risk view. This fragmentation systematically conceals true portfolio exposure and prevents rational risk-adjusted decision-making.

2.2 Behavioral Undiscipline and Emotional Trading

Quantitative studies of retail trading outcomes consistently identify behavioral patterns — not market conditions — as the primary cause of account drawdowns. Patterns such as Revenge Trading (immediately increasing leverage following a loss), Overtrading (high-frequency position entry without edge), and Averaging Down into losing positions are broadly documented. No existing retail platform provides a real-time feedback mechanism that detects and interrupts these patterns as they form. The gap between knowing a behavior is harmful and receiving an objective, in-context warning is the precise space RiskHub occupies.

2.3 Absence of Verifiable Trader Identity and Performance Records

The copy-trading and signal-selling industry is riddled with fraud. Manipulated PnL screenshots, selective track records, and fabricated win-rate statistics are standard practice among fraudulent 'crypto gurus'. Platforms that facilitate copy-trading have no reliable mechanism to verify the true historical risk-adjusted performance of a signal provider. This information asymmetry causes measurable financial harm to thousands of followers who allocate capital to unverified traders. The solution requires a tamper-proof, publicly queryable, and immutable record of a trader's true behavioral discipline and performance metrics — one that cannot be cherry-picked, edited, or deleted.

3. Goals, Objectives & Success Metrics

3.1 Product Goals

- Provide retail crypto traders with a single-pane-of-glass view of their cross-exchange portfolio risk exposure.
- Detect and proactively interrupt emotional and undisciplined trading behaviors in real time.
- Create an immutable, blockchain-based trading identity that transparently communicates a trader's true risk profile.
- Build a trust layer for the copy-trading ecosystem by enabling credential verification via on-chain SBTs.

3.2 MVP Success Metrics (OKRs)

Objective	Key Result	Target (MVP Launch + 90 days)
User Adoption	Registered users with at least 1 connected exchange	500 active users
Engagement	Daily Active Users / Monthly Active Users ratio	>= 35% DAU/MAU
Alert Efficacy	Behavioral alerts acknowledged by user (not dismissed)	>= 60% acknowledgement rate
Web3 Adoption	SBTs minted by users with >= 30 days of trade history	>= 150 SBTs minted
Data Integrity	API key read-only enforcement verified	0 instances of write/withdrawal access
Performance	Dashboard data refresh latency (P95)	< 3 seconds

4. Target Audience & User Personas

4.1 Primary Persona: The Active Retail Trader

Attribute	Description
Name	Alex — The Active Trader
Demographics	25–40 years old, tech-comfortable, trades crypto as a primary or secondary income source
Behaviors	Maintains accounts on 2–5 exchanges simultaneously; actively trades Spot and Futures; checks positions multiple times daily; follows trading communities online
Pain Points	Cannot aggregate total portfolio exposure; has experienced revenge trading patterns; doubts the legitimacy of copy-trade gurus; fears providing API keys to third-party platforms
Goals	Maximize risk-adjusted returns; improve emotional discipline; build a verifiable track record to attract copy-traders; minimize catastrophic drawdowns
Motivation to Use RiskHub	Wants a trusted, read-only tool that helps them understand and manage risk without requiring platform custody of funds

4.2 Secondary Persona: The Copy-Trade Expert / Signal Provider

Attribute	Description
Name	Jordan — The Signal Provider
Behaviors	Shares trading signals or runs copy-trade groups; needs to prove legitimacy to attract followers; uses social media and trading platforms for community building
Pain Points	Followers are skeptical due to widespread fake PnL; no portable, trusted credential for performance; competes with fraudulent actors
Goal	Mint an immutable, public SBT that objectively demonstrates their discipline score, max drawdown, and win rate — shareable as a trust badge

4.3 Tertiary Persona: The Copy-Trade Follower

A risk-averse investor who allocates capital to copy-trade experts but lacks the tools to independently verify performance claims. RiskHub's SBT system provides this user with an on-chain data source that can be queried before committing funds.

5. System Architecture

5.1 Architecture Overview

RiskHub is built on a four-layer architecture designed for high performance, security, and extensibility. Each layer has a defined responsibility boundary and communicates with adjacent layers through well-defined APIs and event streams.

Layer	Technology Stack	Responsibilities
Client Layer	Next.js 14, React, TypeScript, Tailwind CSS, TradingView Lightweight Charts, React Flow, Wagmi, RainbowKit	UI rendering, wallet connection, real-time data display, WebSocket event handling, alert delivery
Server Layer	Python 3.11, FastAPI, CCXT, WebSockets, Celery (task queue)	Exchange API orchestration, quant engine execution, alert generation, Oracle signing, REST + WebSocket API serving
Data Layer	MongoDB (primary), Redis (caching & pub/sub)	Persistent storage of trade history, user profiles, SBT metadata, risk metrics time-series; caching for real-time dashboard data
Web3 / Blockchain Layer	Web3.py, Solidity ^0.8.x, BNB Chain Testnet / Ethereum Sepolia, ERC-5114 / ERC-721	Off-chain Oracle data signing, smart contract interaction, SBT minting, on-chain risk record storage and public query interface

5.2 Data Flow — Read-Only Ingestion Pipeline

User provides read-only API key and secret via the frontend. Credentials are encrypted (AES-256) before storage in MongoDB. The FastAPI backend invokes the CCXT library to fetch balances, positions, and trade history from the target exchange. Fetched data is normalized to a unified schema and stored in MongoDB. A background Celery task re-fetches data on a configurable schedule (default: every 60 seconds for live positions, every 5 minutes for historical trade data). Normalized data is published to a Redis pub/sub channel, which the WebSocket server subscribes to, pushing updates to connected frontend clients.

5.3 Data Flow — Quant Engine & Alert Pipeline

Upon each data update, the Quant Engine is invoked as a Celery task. The engine reads trade history from MongoDB and applies the full suite of heuristic rule evaluators (see Section 6.2). Any triggered rules generate Alert objects, which are persisted to MongoDB and simultaneously pushed via WebSocket to the connected frontend client. The frontend renders alerts as dynamic, non-blocking overlay notifications.

5.4 Data Flow — SBT Minting Pipeline

User initiates SBT mint from the frontend. The backend aggregates the user's complete risk metrics payload (drawdown, win rate, leverage profile, discipline score) and signs it using a private key held by the RiskHub Oracle service (Web3.py). The signed payload and signature are returned to the frontend. The user's MetaMask wallet (connected via Wagmi + RainbowKit) submits a transaction to the RiskHubSBT smart contract, calling the mint function with the signed payload. The smart contract verifies the Oracle's signature, creates a non-transferable SBT bound to the user's wallet address, and emits a SBTMinted event. The token is publicly queryable on-chain by any third party.

5.5 Security Architecture

- Read-Only Enforcement: CCXT library is configured exclusively with read-scope API keys. No trade execution, withdrawal, or fund transfer methods are called or exposed.
- Credential Security: Exchange API keys are encrypted at rest using AES-256 symmetric encryption. Encryption keys are managed via environment secrets (not stored in the database).
- Oracle Key Security: The RiskHub Oracle signing key is stored in a Hardware Security Module (HSM) or secure key vault in production. The key is never exposed to the client layer.
- Authentication: All backend API endpoints require JWT authentication. Web3 wallet signature (Sign-In With Ethereum / Sign-In With BNB) is supported as an alternative auth mechanism.
- Network Security: All client-server communication over HTTPS/WSS. CORS restricted to known frontend origin domains.

6. Detailed Functional Requirements

This section defines each product feature in terms of user stories, detailed behavior, and MVP acceptance criteria.

6.1 Feature: Read-Only Cross-Exchange Data Ingestion

Priority: P0 — MVP Critical

6.1.1 User Story

As an active trader, I want to connect my exchange accounts using read-only API keys so that RiskHub can aggregate my portfolio data without ever having access to execute trades or withdraw funds.

6.1.2 Functional Specifications

- The system shall support API key connection for Binance and OKX in the MVP, with a framework extensible to all 100+ CCXT-supported exchanges in future phases.
- The API key connection interface shall clearly label each permission scope (Read-Only) and block any attempt to store keys with write or withdrawal permissions.
- The system shall fetch and unify the following data points per connected exchange: total account balance (USDT equivalent), Spot holdings (asset, quantity, current value), active Futures positions (symbol, side, entry price, current price, leverage, unrealized PnL, liquidation price, margin used), and historical closed trades (timestamp, symbol, side, quantity, entry price, exit price, realized PnL, leverage used).
- All data from multiple exchanges shall be normalized to a single internal schema and aggregated into a unified portfolio view, denominated in USDT.
- The system shall re-sync live position data every 60 seconds and historical trade data every 5 minutes.
- If an exchange API key becomes invalid or returns an error, the system shall surface a non-blocking warning notification to the user without crashing the dashboard.

6.1.3 MVP Acceptance Criteria

1. A user can successfully add a Binance read-only API key and see their Spot and Futures portfolio data rendered on the dashboard within 10 seconds of connection.
2. A user can successfully add an OKX read-only API key, and data from both exchanges is aggregated and displayed in a single unified balance view.
3. The system is demonstrably incapable of placing any trade or withdrawal on a connected exchange account (verified via CCXT configuration audit).
4. An invalid or expired API key triggers a user-visible warning badge on the exchange connection panel within one refresh cycle.

5. Portfolio balance values update without requiring a manual page refresh (WebSocket-driven updates confirmed).

6.2 Feature: Behavioral Quant Engine & Dynamic Risk Alerts

Priority: P0 — MVP Critical

6.2.1 User Story

As a trader prone to emotional reactions, I want to receive real-time, context-aware warnings when my trading behavior matches patterns associated with emotional or undisciplined trading, so that I can pause, reconsider, and protect my capital.

6.2.2 Behavioral Rules Engine — Rule Definitions

The following rules constitute the MVP ruleset. Each rule is evaluated after every data sync cycle.

Rule ID	Behavior Pattern	Trigger Condition	Alert Message
RQ-001	Revenge Trading	Loss > 5% equity AND new position opened within 10 mins with leverage >= 15x	WARNING: Revenge Trading Detected. You opened a high-leverage position shortly after a significant loss. Consider stepping away.
RQ-002	Overtrading	> 10 trades opened within a 60-minute window	NOTICE: Overtrading Pattern Detected. High-frequency position opening may indicate emotional trading. Review your trade log.
RQ-003	Excessive Leverage	Any single position opened with leverage > 50x	HIGH RISK: Extreme Leverage Detected. A position using 50x or higher leverage creates severe liquidation risk.
RQ-004	Max Drawdown Breach	Portfolio drawdown from peak exceeds 20%	ALERT: Maximum Drawdown Threshold Breached. Your portfolio has declined 20% from its recent peak. Review position sizing.
RQ-005	Concentration Risk	Single asset constitutes > 60% of total portfolio value	RISK: High Portfolio Concentration. Over 60% of your portfolio is in a single asset. Consider rebalancing.

6.2.3 Calculated Metrics

- Max Drawdown: Calculated as ((Peak Portfolio Value - Trough Portfolio Value) / Peak Portfolio Value) * 100, computed across the full available historical trade window.
- Trading Discipline Score (TDS): A composite score (0–100) derived from the inverse frequency of triggered behavioral rule violations, normalized by trading volume. A score of 100 indicates zero behavioral violations. Score components are: Leverage Control (30%), Drawdown Management (30%), Trade Frequency Discipline (20%), Loss Recovery Behavior (20%).
- Win Rate: (Number of profitable closed trades / Total closed trades) * 100.
- Average Leverage: Mean leverage across all closed Futures positions in the trailing 30-day window.

6.2.4 Alert Delivery Requirements

- Alerts shall be delivered as dynamic overlay pop-up cards in the bottom-right of the dashboard, persisting until acknowledged by the user.
- Each alert card shall display: severity icon (color-coded: red = high, orange = medium, yellow = info), alert title, detailed description message, triggered timestamp, and an action button (Acknowledge / View Details).
- Acknowledged alerts are archived in a persistent Alert History panel accessible from the dashboard sidebar.
- Alerts shall be delivered via WebSocket push; no polling mechanism shall be used for alert delivery.

6.2.5 MVP Acceptance Criteria

6. The Quant Engine correctly detects a simulated Revenge Trading event (RQ-001) and delivers the corresponding alert to the frontend WebSocket within 5 seconds of the trigger event.
7. The Trading Discipline Score renders correctly on the dashboard and updates after each data sync cycle.
8. Max Drawdown is calculated correctly for a test dataset and matches a pre-verified expected output (within 0.01% tolerance).
9. Alert cards render correctly, persist until acknowledged, and are archived in the Alert History panel upon acknowledgement.
10. All five MVP behavioral rules (RQ-001 through RQ-005) are implemented and unit-tested with pass/fail test cases.

6.3 Feature: Asset Contagion Graph

Priority: P1 — MVP Important

6.3.1 User Story

As a portfolio manager, I want to visually understand the correlations between my held assets so that I can identify when my portfolio is effectively concentrated in a single risk factor and proactively reduce cascading liquidation exposure.

6.3.2 Functional Specifications

- The Contagion Graph shall be rendered using React Flow as an interactive, zoomable, and pannable network visualization embedded on the main dashboard.
- Each asset in the user's portfolio shall be represented as a node. Node size shall be proportional to the asset's weight as a percentage of total portfolio value.
- Edges between nodes shall represent pairwise Pearson correlation coefficients calculated from 30-day rolling daily returns. Edge thickness shall be proportional to correlation magnitude (|r|). Edges with |r| < 0.3 shall not be displayed to reduce visual noise.
- Node and edge color coding: Correlation >= 0.7 (high): Red edge; Correlation 0.4–0.69 (moderate): Orange edge; Correlation < 0.4 (low): Grey edge. If a node's asset drops more than 5% in a single day, the node and all connected edges with |r| >= 0.5 shall pulse red to indicate contagion risk.
- Hovering over any node shall display a tooltip showing: asset name, current value, portfolio weight (%), and a list of the top 3 most correlated assets with their correlation coefficient values.
- A Contagion Risk Score shall be calculated as the average correlation of the top-3 asset pairs by portfolio weight. This scalar score (0–100) shall be displayed prominently above the graph.

6.3.3 MVP Acceptance Criteria

11. The graph correctly renders nodes for all assets in a test portfolio with more than 3 holdings.
12. Node sizes are proportionally correct (verified by visual inspection against known portfolio weights).
13. Correlation values are computed correctly against a pre-verified reference dataset (within 0.02 tolerance).
14. A simulated 6% single-day drop in BTC triggers the visual contagion pulse effect on all BTC-adjacent nodes with correlation >= 0.5.
15. The graph is interactive: zoom, pan, and hover tooltip all function correctly in Chromium-based browsers.

6.4 Feature: Web3 Immutable Trading Identity (SBT Minting)

Priority: P1 — MVP Important

6.4.1 User Story

As a signal provider, I want to mint a Soulbound Token that permanently encodes my verified trading discipline metrics on the blockchain so that potential copy-trade followers can independently verify my performance history without relying on screenshots or self-reported data.

6.4.2 Smart Contract Specifications

- Contract Name: RiskHubSBT
- Standard: ERC-5114 (Soulbound) with ERC-721 view interface for compatibility with explorers
- Network (MVP): BNB Chain Testnet; Production-ready: BNB Chain Mainnet and Ethereum Mainnet
- Token Transferability: MUST be non-transferable. The transfer function SHALL revert with error SBTNotTransferable() on any transfer attempt.
- Minting Restriction: Each wallet address is permitted exactly one SBT (one-per-wallet enforcement). Existing SBTs may be updated (not replaced) by re-calling the mint/update function with a fresh Oracle-signed payload.
- Oracle Verification: The contract SHALL verify that the data payload is signed by the canonical RiskHub Oracle address. Minting with an invalid or unauthorized signature SHALL revert.

6.4.3 On-Chain Data Payload

Field	Type	Description
walletAddress	address	The holder's Ethereum-compatible wallet address
maxDrawdown	uint16	Historical max drawdown in basis points (e.g., 1500 = 15.00%)
winRate	uint16	Win rate in basis points (e.g., 5800 = 58.00%)
avgLeverage	uint8	Average leverage multiplier across all Futures positions
disciplineScore	uint8	Trading Discipline Score (0–100)
totalTrades	uint32	Cumulative number of closed trades at time of minting
dataTimestamp	uint64	Unix timestamp of data aggregation by the Oracle
oracleSignature	bytes	ECDSA signature by RiskHub Oracle over the keccak256 hash of the above fields

6.4.4 Frontend Minting Flow

16. User navigates to the Risk ID / SBT panel on the dashboard.

17. System displays current calculated metrics (drawdown, win rate, discipline score) and an estimated gas fee for the minting transaction.
18. User clicks Mint My Risk ID. The backend Oracle signs the payload and returns the signature.
19. MetaMask (or compatible wallet via RainbowKit) presents the transaction confirmation dialog to the user.
20. Upon on-chain confirmation, the dashboard updates to display the minted SBT with a Verified badge, token ID, and block explorer link.

6.4.5 MVP Acceptance Criteria

21. The RiskHubSBT contract is deployed to BNB Chain Testnet and passes all unit tests including: mint with valid Oracle signature succeeds; mint with invalid signature reverts; transfer attempt reverts with SBTNotTransferable error; second mint from same wallet correctly updates existing SBT metrics.
22. A user with >= 30 days of connected trade history can complete the full minting flow (backend signing + wallet transaction) within the dashboard.
23. The minted SBT is queryable on BNB Chain Testnet block explorer with correctly populated on-chain data fields.
24. The frontend displays the SBT verification badge and block explorer link upon successful mint confirmation.

7. UI/UX Requirements

7.1 Design Language

- Color Palette: Dark-mode primary. Background: #0F172A (deep navy). Surface: #1E293B. Accent: #1A56DB (electric blue). Success: #10B981. Warning: #F59E0B. Danger: #EF4444.
- Typography: System-native sans-serif stack (Inter preferred). Financial metrics rendered in monospace for alignment.
- All charts and graphs shall use TradingView Lightweight Charts (financial data) and React Flow (network graph). No third-party chart libraries shall introduce unnecessary bundle size.
- The layout shall be fully responsive for desktop (min 1280px) and tablet (min 768px) breakpoints. Mobile view is out of scope for MVP.

7.2 Core Dashboard Layout

- Top Navigation Bar: Logo, connected wallet address (truncated), global portfolio value (large, prominent), last sync timestamp, and notification bell icon.
- Left Sidebar: Navigation links (Dashboard, Risk Analysis, SBT Identity, Alert History, Settings). Exchange connection status badges per connected exchange.
- Main Content Area (3-column grid):
  - Column 1 (narrow): Portfolio breakdown card — unified balance, per-exchange breakdown, Discipline Score gauge.
  - Column 2 (wide): Asset Contagion Graph, Contagion Risk Score.
  - Column 3 (narrow): Open Positions list, Max Drawdown indicator, Recent Alerts panel.
- Alert Overlay: Dynamic, non-blocking alert cards in the bottom-right corner, with severity color coding and dismiss controls.

8. Non-Functional Requirements

Category	Requirement	Specification
Performance	Dashboard Load	Initial page load (First Contentful Paint) < 2.5s on standard broadband. WebSocket data refresh latency P95 < 3s.
Performance	Quant Engine	Behavioral rule evaluation pipeline completes in < 500ms per user per sync cycle.
Scalability	Concurrent Users	System shall support 1,000 concurrent authenticated users with active WebSocket connections without degradation. Horizontally scalable via stateless FastAPI workers.
Reliability	Uptime	99.5% monthly uptime SLA for all production services. Exchange API failures shall be handled gracefully with retry logic (exponential backoff, 3 retries).
Security	Data Encryption	All API keys encrypted at rest (AES-256). All data in transit encrypted (TLS 1.3+). No sensitive credentials logged.
Security	Smart Contract	Smart contract shall be audited by a third-party security firm before Mainnet deployment. Testnet deployment requires internal security review.
Maintainability	Code Quality	Backend test coverage >= 80% (unit + integration). Frontend component test coverage >= 60%. CI/CD pipeline required for all merges to main.
Compliance	Data Privacy	User trade data shall not be shared with or sold to third parties. Privacy policy must clearly disclose data retention and deletion rights.
Observability	Monitoring	Application-level metrics (latency, error rate, alert trigger rate) shall be exposed via a Prometheus-compatible metrics endpoint. Structured JSON logging required on all backend services.

9. MVP Scope Definition

9.1 In Scope for MVP

IN SCOPE	OUT OF SCOPE (Post-MVP)
• Binance + OKX read-only API integration • Unified portfolio dashboard (dark mode) • 5 behavioral alert rules (RQ-001 to RQ-005) • Trading Discipline Score calculation • Asset Contagion Graph (React Flow) • Max Drawdown & Win Rate metrics • SBT minting on BNB Chain Testnet • MetaMask wallet connection (RainbowKit) • Alert history panel & acknowledgement flow • JWT-based user authentication	• Mobile application (iOS / Android) • ML-based behavioral prediction models • Third-party copy-trade platform API integration • Mainnet SBT deployment • Social sharing / SBT public profile page • Decentralized exchange (DEX) data integration • In-app notifications (email, push) • Team / multi-user accounts • Fiat on/off ramp integration • Customizable alert rule builder (user-defined rules)

10. Delivery Plan & Sprint Milestones

The MVP is scoped for a 16-week delivery cycle across four two-sprint phases (2-week sprints).

Phase	Weeks	Deliverables	Exit Criteria
Phase 1: Foundation	1–4	Backend scaffolding (FastAPI), MongoDB schema design, CCXT integration for Binance, user authentication (JWT), frontend Next.js scaffold, Tailwind design system	Binance read-only data successfully fetched and stored. User login flow working end-to-end.
Phase 2: Core Dashboard	5–8	OKX integration, unified portfolio dashboard UI, WebSocket real-time updates, Contagion Graph (React Flow) with correlation calculation, exchange connection management UI	Dashboard displays live data from both exchanges. Contagion graph renders with correct correlation-based edge weights.
Phase 3: Quant Engine	9–12	Full Behavioral Rules Engine (RQ-001 to RQ-005), Discipline Score calculation, Max Drawdown engine, WebSocket alert push, Alert History panel, alert card UI component	All 5 behavioral rules trigger correctly in integration tests. Alerts delivered via WebSocket within 5s of trigger.
Phase 4: Web3 & Launch	13–16	Solidity smart contract development + testing, Oracle signing service (Web3.py), RainbowKit wallet integration, SBT minting UI flow, BNB Chain Testnet deployment, end-to-end QA, security review, soft launch preparation	SBT minting completes successfully on Testnet. All MVP acceptance criteria passed. No critical security findings.

11. Risks, Assumptions & Dependencies

11.1 Key Risks

Risk	Severity	Mitigation
Exchange API rate limits impacting real-time data freshness	Medium	Implement request queuing with rate-limit-aware CCXT configuration; use Redis caching to serve stale-but-recent data during rate limit windows
User distrust of providing API keys to a third-party platform	High	Prominent security messaging; open-source the CCXT integration layer; consider an optional local-only mode for ultra-paranoid users in Post-MVP
Smart contract vulnerability leading to reputation damage pre-Mainnet	High	Mandatory third-party audit before Mainnet. Testnet deployment with limited user group for extended soak testing
Behavioral alert false positives eroding user trust	Medium	Conservative threshold calibration for MVP. User feedback mechanism on each alert (Was this helpful?). Iterative threshold refinement post-launch
Regulatory classification of SBT as a financial instrument	Low-Med	Legal review of SBT as an informational credential (not a security); ensure no token carries monetary value or ownership rights

11.2 Assumptions

- Exchange APIs (Binance, OKX) remain stable and do not introduce breaking changes to their read-only API schema during the MVP build cycle.
- BNB Chain Testnet remains operational and publicly accessible throughout the development and testing phases.
- Users possess basic familiarity with MetaMask or equivalent EVM-compatible browser wallets for SBT minting.
- The product team has access to test accounts on both Binance and OKX with documented trade history for integration testing.

11.3 External Dependencies

Dependency	Owner	Impact if Unavailable
CCXT Library	Open Source (MIT)	Core data ingestion blocked; would require proprietary exchange integration work
Binance / OKX APIs	Binance / OKX	Affected exchange's data unavailable; dashboard falls back to last-known data with error badge
BNB Chain Testnet RPC	BNB Chain Foundation	SBT minting testing blocked; switch to Ethereum Sepolia as fallback

TradingView Lightweight Charts	TradingView (Open Source)	Financial chart visualizations degraded; fallback to recharts

12. Open Questions & Decisions Required

#	Open Question	Owner	Target Resolution
1	Should users be required to connect a wallet address before accessing the dashboard, or can they use the platform without Web3 features?	Product + Engineering	Sprint 2
2	What is the precise AES-256 key management strategy for API key encryption in the MVP environment (KMS vs. environment secrets)?	Engineering + Security	Sprint 1
3	Should behavioral alert thresholds (e.g., the 5% loss threshold for RQ-001) be user-configurable in MVP, or fixed globally?	Product	Sprint 3
4	Will the SBT token metadata (IPFS vs. on-chain) be stored fully on-chain for MVP, or will non-critical fields be offloaded to IPFS?	Engineering + Blockchain	Sprint 4
5	What is the legal entity structure for operating the Oracle signing service, and does it create any custodial or financial service obligations under applicable regulations?	Legal	Pre-Launch

13. Glossary of Terms

Term	Definition
CCXT	CryptoCurrency eXchange Trading Library. An open-source library providing a unified API interface to 100+ cryptocurrency exchanges.
SBT (Soulbound Token)	A non-transferable, non-tradeable blockchain token permanently bound to a specific wallet address, representing verifiable credentials or identity attributes.
Max Drawdown	The maximum observed percentage decline in portfolio value from a historical peak to a subsequent trough, before a new peak is achieved.
Revenge Trading	The behavioral pattern where a trader immediately re-enters the market with increased leverage or position size following a significant loss, driven by the emotional desire to recover losses quickly.
Oracle	In the Web3 context, a trusted off-chain service that fetches real-world data, signs it cryptographically, and delivers it to a smart contract in a verifiable format.
Contagion Risk	The risk that a decline in one asset triggers correlated declines in other held assets due to high cross-asset correlation, amplifying total portfolio losses beyond what any single position would suggest.
Trading Discipline Score (TDS)	A proprietary composite metric (0–100) calculated by RiskHub's Quant Engine reflecting the consistency of a trader's adherence to risk management best practices over time.
ERC-5114	An Ethereum Improvement Proposal standard defining the interface for Soulbound tokens — non-transferable NFTs bound to a specific EOA or smart contract wallet.
P95 Latency	The 95th percentile response time. 95% of all requests are completed within this time threshold. A standard metric for measuring system performance under real-world load.
Read-Only API Key	An exchange API credential with permissions restricted exclusively to reading account data (balances, positions, trade history). Cannot execute trades, place orders, or trigger withdrawals.

14. Approval & Sign-Off

This document requires review and approval from all listed stakeholders prior to development commencement. Approved signatures constitute authorization to proceed with the MVP scope as defined herein.

Name & Title	Role	Signature	Date

	Product Lead		
	Engineering Lead		
	Blockchain Lead Design Lead		
	Legal Counsel		